

Discussion section 5

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1 Chp 3, Q15

On a chessboard a single random knight performs a simple random walk. From any square, the knight chooses from among its permissible moves with equal probability. If the knight starts on a corner, how long, on average, will it take to return to that corner?

For your convenience, chessboard is 8×8 , and a knight moves in an L-shape: 2 squares in one direction (horizontal or vertical), then 1 square perpendicular to that.

Proof. This is a simple random walk on the graph. By textbook, page 91, the stationary distribution for the vertex v ,

$$\pi_v = \frac{\deg(v)}{\sum_z \deg(z)}.$$

For the knight, it has the following graph. The number on each square stands for the degree of that square. For example, to reach top left corner there are only two ways: from the red square, going left, going left, and going up; or from the orange square, going up, going up, and going left. However in the middle region you have 8 ways to access.

2	3	4	4	4	4	3	2
3	4	6	6	6	6	4	3
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
4	6	8	8	8	8	6	4
3	4	6	6	6	6	4	3
2	3	4	4	4	4	3	2

Then for a corner c , $\pi_c = \frac{2}{4 \cdot 2 + 8 \cdot 3 + 20 \cdot 4 + 16 \cdot 6 + 16 \cdot 8}$, and (perhaps with some thoughts) this is an irreducible finite Markov chain, so the expected returning time is $\mu_c = 1/\pi_c = 168$. $\square$

2 Chp 3, Q16

As in the previous exercise, find the expected return time from a corner square for the following chess pieces: (i)queen,(ii)rook,(iii)king,(iv)bishop. Order the pieces by which pieces return quickest. For your interest,

(i)A queen moves to any number of vacant squares in a straight line: vertically, horizontally, or diagonally.

(ii)A rook moves to any other square in the same row/column.

(iii)A king moves to adjacent squares.

(iv)A bishop moves diagonally any number of open squares in a straight line.

Proof. (i) For the queen, it has the following graph:

	21	21	21	21	21	21	21	21
	21	23	23	23	23	23	23	21
	21	23	25	25	25	25	23	21
	21	23	25	27	27	25	23	21
	21	23	25	27	27	25	23	21
	21	23	25	25	25	25	23	21
	21	23	23	23	23	23	23	21
	21	21	21	21	21	21	21	21

For a corner c , $\pi_c = \frac{21}{4 \cdot 27 + 12 \cdot 25 + 20 \cdot 23 + 28 \cdot 21}$, and clearly it's an irreducible finite Markov chain, so the expected returning time is $\mu_c = 1/\pi_c = 1456/21 \approx 69.33..$

(ii) For the rook, it has the following graph:

	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14
	14	14	14	14	14	14	14	14

For a corner c , $\pi_c = \frac{14}{14 \times 64}$, and clearly it's an irreducible finite Markov chain, so the expected returning time is $\mu_c = 1/\pi_c = 64$.

(iii) For the king, it has the following graph:

	3	5	5	5	5	5	5	3
	5	8	8	8	8	8	8	5
	5	8	8	8	8	8	8	5
	5	8	8	8	8	8	8	5
	5	8	8	8	8	8	8	5
	5	8	8	8	8	8	8	5
	5	8	8	8	8	8	8	5
	5	8	8	8	8	8	8	5
	3	5	5	5	5	5	5	3

For a corner c , $\pi_c = \frac{3}{4 \cdot 3 + 24 \cdot 5 + 36 \cdot 8} = \frac{3}{420}$, and clearly it's an irreducible finite Markov chain, so the expected returning time is $\mu_c = 1/\pi_c = 140$.

(iv) For the bishop, it has the following graph:

	7	7	7	7	7	7	7	7
	7	9	9	9	9	9	9	7
	7	9	11	11	11	11	9	7
	7	9	11	13	13	11	9	7
	7	9	11	13	13	11	9	7
	7	9	11	11	11	11	9	7
	7	9	9	9	9	9	9	7
	7	7	7	7	7	7	7	7

For a corner c , $\pi_c = \frac{7}{4 \cdot 13 + 12 \cdot 11 + 20 \cdot 9 + 28 \cdot 7} = \frac{7}{560}$, and clearly it's an irreducible finite Markov chain, so the expected returning time is $\mu_c = 1/\pi_c = 560/7 = 80$.

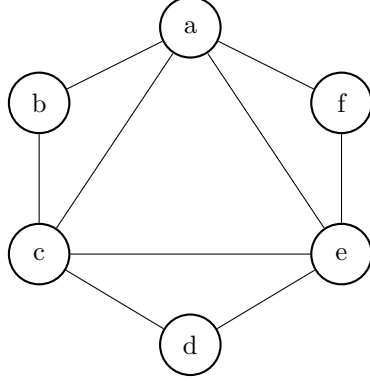
Thus, the first returning times are ordered as:

$$T_{rook} < T_{queen} < T_{bishop} < T_{king}.$$

□

3 Chp 3, Q19

Consider random walk on the graph. Use first-step analysis to find the expected time to hit d for the walk started in a . (Hint: By exploiting symmetries in the graph, the solution can be found by solving a 3×3 linear system.)



Proof. Let $e_x = \mathbb{E}[T_d | X_0 = x]$, where T_d is the first passage time to the state d . The first step analysis yields,

$$\begin{cases} e_a = \frac{1}{4}(1 + e_b) + \frac{1}{4}(1 + e_c) + \frac{1}{4}(1 + e_e) + \frac{1}{4}(1 + e_f), \\ e_b = \frac{1}{4}(1 + e_a) + \frac{1}{4}(1 + e_c), \\ e_c = \frac{1}{4}(1 + e_b) + \frac{1}{4}(1 + e_a) + \frac{1}{4}(1 + e_e) + \frac{1}{4}. \end{cases}$$

(Obviously we omit the equations for state f and state e since we won't use them.)

By symmetry of the graph we have $e_c = e_e$ and $e_b = e_f$. Thus the above is reduced to a system with three unknowns e_a , e_b and e_c :

$$\begin{cases} e_a = \frac{1}{2}(1 + e_b) + \frac{1}{2}(1 + e_c), \\ e_b = \frac{1}{4}(1 + e_a) + \frac{1}{4}(1 + e_c), \\ e_c = \frac{1}{4}(1 + e_b) + \frac{1}{4}(1 + e_a) + \frac{1}{4}(1 + e_c) + \frac{1}{4}. \end{cases}$$

Thus $e_a = e_b = 10$, $e_c = 8$. The expected time to hit d for the walk started in a is $e_a = 10$. $\square$